Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

The main concepts might be used in the project are helper function writing, nested dataclass, and the drafter program. I expect to use conditionals to check if the text or input user giving is valid and classes to clarify each compound and their attribute. I would like to improve drafter program writing skills because I am not actually familiar with using this skill.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I would say the hardest thing is to think about how this whole program works and organize the dataclass, variable and helper function I may use. The easiest part for me is to write a helper function. After I have an overall logical diagram of the whole program in my mind, I can focus on writing each individual function. It may be easy for me when I have a clear idea about what the function should do and how it works.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Debugging the program may require the most of my time and effort. This program requires many functions and variables. It's hard to not make mistakes during programming. So, I will try to work the program page by page. After a page successfully works, I go to write or debug the next page. It’s important for the whole program that it should be implemented first.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Probably no interference. Writing this big program at once is hard for me which makes me distracted. So, I will separate this whole program by dataclass writing, helper each function writing and debugging. This strategy may help me finish the work efficiently and also in high-quality.

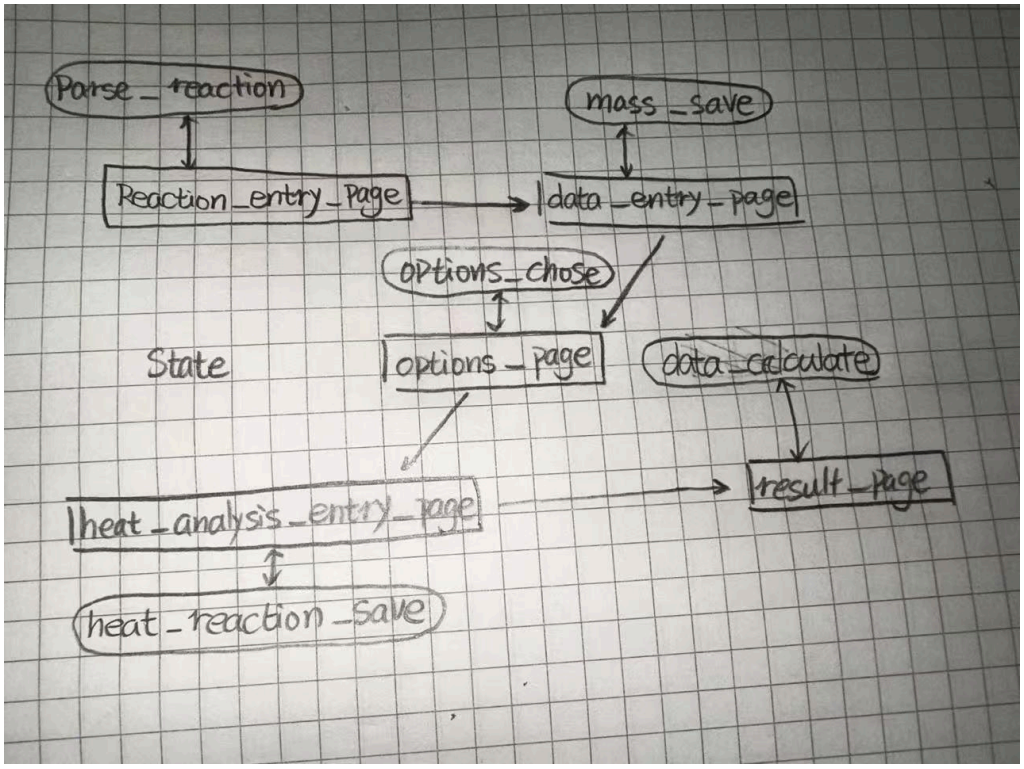
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The main purpose of the site is to help people to calculate some chemical value without calculation which is a helpful tool for users.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second	**substances: list[Substance]** substances represents a list of substances used in reaction which is given by the user. Putting them into a list is easy to get each molar mass from ChemPy. **show_heat_analysis: bool**

dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	This attribute is to decide whether the user wants to get a heat analysis for reaction. If it is False, the result page will only show the simple chemical value. If True, the site will require a heat of reaction from users and give them answers. **limiting_reagent: str** **theorietical_yield: str** **remaining_masses: list[str]** **heat_analysis: str** **percent_yield: str** These five attributes just represent the final value the result page will show to the users.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	My second dataclass is called Substance which helps the site to save different values like molar mass of each substance the user gave. Substance: actual_mass: float molar_mass: float is_product: bool is_reactant: bool coefficient: int
Routes (minimum 7) The names of your route functions	parse_reaction mass_save options_chose reaction_entry_page data_entry_page options_page heat_analysis_entry_page result_page
Pages (minimum 5) A brief description of the pages in your site	Page 1: Asking users input a balanced chemical reaction Page 2: Asking users input each mass of reactants and the actual mass of product that the reaction

	produces. Page 3: Users can choose what specific data they want to get about this reaction. Page 4: Asking users if they want a heat analysis. Page 5: the result page that shows all the value and calculation results the users want.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox1: It is on the first page for users to enter a balanced chemical reaction. TextBox2/3/4/5: It is on the second page for users to enter the masses of reactants and the actual masses of products that are produced. SelectBox1/2/3/4: Asking users what specific data they want. Including limiting reactant, theoretical yield and remaining masses of reactants. (Page 3) CheckBox1: It is on the fourth page for users to choose if they want a heat analysis. TextBox6: Enter a heat of reaction(Page 4)
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	I will use a lot of if statements in this site. The main use is to check the text or number is valid, to clarify the type of substances, and to confirm what users want.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	A loop for checking if the users enter valid information. If they enter invalid information, the textbox will be cleared and returned to a new blank page. It means checking the information and making sure the whole program works.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Question:
Hi, I am confused about how to define attributes of class State when programming a drafter. Can you explain what state is and how it is used? Also, I have another dataclass called Compound in my site, how can I distinguish between these two dataclass variables?

Answer:
We can set aside those complicated computer terms and use an everyday analogy:

1. **What is State? (Imagine it as a "backpack")** When building a web page, the page refreshes and changes. User inputs, clicked buttons, and calculation results all need somewhere to be stored, right?
2.
    - State is just a "backpack."
    -
    - The Drafter framework requires that all data while the web page is running (such as the equation the user entered or the calculated results) must be packed into this backpack.
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    - As long as they are put in the backpack, any part of the web page can take them out and use them at any time.
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3. **What is a Compound? (Imagine it as an "ID card" for each chemical substance)** Your chemical reaction usually has several substances (like hydrogen, oxygen, water).
4.
    - Every substance has its own name, stoichiometric coefficient, and mass.
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    - Without classes, you might need to write a bunch of messy, unorganized lists. Writing a Compound (ID card) is equivalent to setting a template for every substance: as long as it's a chemical substance, it just fills in the

blanks according to this template.

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- Once you have these "ID cards," when writing code later, you can use a simple loop to print or calculate all of them.
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If writing this object-oriented code feels a bit tough at first, we can break the task down into even smaller pieces.

Looking at this now, do you find Python's syntax (like `@dataclass`) unfamiliar, or is the logic of "how web pages link up with code" a bit hard to understand?

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.